Division of Work

Energy-Aware Adaptive CPU Scheduling for Heterogeneous Multicore Edge Devices Using Deep Reinforcement Learning

Final submission | 26 October 2026 | Repository: https://github.com/AdityaRezan/energy-aware-drl-scheduler

Companion to Energy_Aware_CPU_Scheduling_Final_38_Pages.pdf. Section 28 of the review gives the functional summary; this document records the detailed allocation across the five team members.

1. Team roster

#	Name	Reg. No.	Institutional email	Primary role
1	Aditya Rajeshwar Karn	24BEL0087	aditya.rajeshwar2024@vitstudent.ac.in	Team lead - environment, training and final integration
2	Yuvraj Singh	24BEL0057	yuvraj.singh2024@vitstudent.ac.in	C inference, weight export and parity evidence
3	Aditya Singh	24BEL0008	aditya.singh2024c@vitstudent.ac.in	Native schedulers, C engine and the test suite
4	Atharva Sivaraman	24BEL0082	atharva.sivaraman2024@vitstudent.ac.in	Evaluation, statistics, reproducibility and CI
5	Yash Jain	24BEL0035	yash.jain2024a@vitstudent.ac.in	Documentation, dashboard and document editor

2. Division of the 38-page literature review

The review is divided into five contiguous parts aligned with section boundaries. Each part has one author responsible for its accuracy, citations and figures, and one internal reviewer. Pages 36-38 are shared. Page counts include the front matter and the annotated bibliography.

Part	Author	Pages	Sections	Pp.	Scope
I	Aditya Rajeshwar Karn	1-8	Front matter (title, abstract, contents); Sections 1-6	8	Review scope, problem framing, energy semantics, hardware realism, and the 2020-2022 foundations
II	Yuvraj Singh	9-15	Sections 7-13	7	Runtime power learning, thermal-aware DRL, temporal governors, DAG scheduling, multi-deadline DVFS and invocation economics
III	Aditya Singh	16-22	Sections 14-20	7	Joint heterogeneous control, partial observability, HiDVFS, repository architecture, implemented state/actions/reward and local C checks
IV	Atharva Sivaraman	23-29	Sections 21-25	7	Revision priorities, evaluation plan, synthesis, extended review methodology, evidence grading and comparative evidence synthesis
V	Yash Jain	30-35	Sections 26-29; Appendices A and B	6	Remediation and re-measurement, statistical protocol, team governance, limitations and ethics, artefact manifest and glossary

Shared part VI: pages 36-38, Annotated references [1]-[25] (3 pages). Three pages of annotated references are a shared responsibility. Each member verifies five entries for publication status, DOI and role statement. Yash Jain consolidates the bibliography, but every member signs off on their own five.

3. Reference verification split (pages 36-38)

Member	Reg. No.	References	Responsibility
Aditya Rajeshwar Karn	24BEL0087	[1]-[5]	Confirm publication status, DOI/URL and the stated role of each source
Yuvraj Singh	24BEL0057	[6]-[10]	Confirm publication status, DOI/URL and the stated role of each source
Aditya Singh	24BEL0008	[11]-[15]	Confirm publication status, DOI/URL and the stated role of each source
Atharva Sivaraman	24BEL0082	[16]-[20]	Confirm publication status, DOI/URL and the stated role of each source

Member	Reg. No.	References	Responsibility
Yash Jain	24BEL0035	[21]-[25]	Confirm publication status, DOI/URL and the stated role of each source

4. Technical and code ownership

Code ownership follows the functional areas in Section 28.2 of the review. Owning a file means reviewing changes to it, keeping it reproducible, and answering questions about it during the viva.

Area	Owner	Artefacts owned
Environment, training and final integration	Aditya Rajeshwar Karn	edge_env.py, train_model.py, evaluate_model.py, compare_adaptive_policies.py, diagnose_policy.py, inspect_policy_grid.py, trace_policy.py
C inference, weight export and parity evidence	Yuvraj Singh	inference.c, inference.h, weights.h, export_weights.py, test_inference.c, test_policy.py, test_policy_parity.py
Native schedulers, c engine and the test suite	Aditya Singh	main.c, hardware.c, hardware.h, scheduler.c, scheduler.h, task_queue.c, task_queue.h, task.h, drl_scheduler.c, drl_scheduler.h, test_*.c, Makefile
Evaluation, statistics, reproducibility and ci	Atharva Sivaraman	benchmark_equal_work.py, compare_all_policies.c, compare_policies.c, results/equal_work_benchmark.json, .github/workflows/ci.yml
Documentation, dashboard and document editor	Yash Jain	README.md, TEAM.md, CONTRIBUTION.md, CHANGELOG.md, LICENSE, docs/, app.py

5. Deliverables outside the manuscript

Task	Owner	Expected output
Git workflow, builds, tags and release	Aditya Singh	Makefile, CI workflow, clean reproducible build
PPO training, weight export and Python-C parity	Yuvraj Singh	Saved model archives, weights.h, parity log
Benchmark sweep, seeds and machine-readable results	Atharva Sivaraman	results/equal_work_benchmark.json, 20-profile CI
Streamlit dashboard and live demonstration	Yash Jain	app.py, screenshots, demo script
Abstract, final integration and submission package	Aditya Rajeshwar Karn	Final 38-page PDF/DOCX and repository release
Reference verification (five entries each)	All five members	Checked bibliography, pages 36-38

6. Internal peer review

Parts were reviewed round-robin so that no member graded their own work. The reviewer checked the argument, the citation trail and the claim boundaries before the final merge.

Author	Part authored	Reviewed by	Part reviewed
Aditya Rajeshwar Karn	Part I (p. 1-8)	Part II	Yuvraj Singh
Yuvraj Singh	Part II (p. 9-15)	Part III	Aditya Singh
Aditya Singh	Part III (p. 16-22)	Part IV	Atharva Sivaraman
Atharva Sivaraman	Part IV (p. 23-29)	Part V	Yash Jain
Yash Jain	Part V (p. 30-35)	Part I	Aditya Rajeshwar Karn

7. Presentation slots (20-minute talk + 5-minute Q&A)

Slot	Time	Presenter	Topic
1	0:00-4:00	Aditya Rajeshwar Karn	Introduction, motivation, scope and contributions
2	4:00-8:00	Yuvraj Singh	Literature 2022-2024 and the Python-to-C export path
3	8:00-12:00	Aditya Singh	Repository architecture, C engine and baseline schedulers
4	12:00-16:00	Atharva Sivaraman	Evaluation design, statistics and the work-normalised results
5	16:00-20:00	Yash Jain	Remediation, claim boundaries, limitations and conclusion

Slot	Time	Presenter	Topic
-	20:00-25:00	All five members	Questions and answers; the lead moderates and each member answers for their own part

8. Individual responsibility cards

1. Aditya Rajeshwar Karn (24BEL0087)

Email: aditya.rajeshwar2024@vitstudent.ac.in
Primary role: Team lead - environment, training and final integration
Authored: Part I, pages 1-8 (8 pages) - Front matter (title, abstract, contents); Sections 1-6
Focus: Review scope, problem framing, energy semantics, hardware realism, and the 2020-2022 foundations
Owns: edge_env.py, train_model.py, evaluate_model.py, compare_adaptive_policies.py, diagnose_policy.py, inspect_policy_grid.py, trace_policy.py
References: [1]-[5]
Presents: 0:00-4:00 - Introduction, motivation, scope and contributions
Reviews: Part II (Yuvraj Singh)
Indicative effort: about 30 hours

2. Yuvraj Singh (24BEL0057)

Email: yuvraj.singh2024@vitstudent.ac.in
Primary role: C inference, weight export and parity evidence
Authored: Part II, pages 9-15 (7 pages) - Sections 7-13
Focus: Runtime power learning, thermal-aware DRL, temporal governors, DAG scheduling, multi-deadline DVFS and invocation economics
Owns: inference.c, inference.h, weights.h, export_weights.py, test_inference.c, test_policy.py, test_policy_parity.py
References: [6]-[10]
Presents: 4:00-8:00 - Literature 2022-2024 and the Python-to-C export path
Reviews: Part III (Aditya Singh)
Indicative effort: about 28 hours

3. Aditya Singh (24BEL0008)

Email: aditya.singh2024c@vitstudent.ac.in
Primary role: Native schedulers, C engine and the test suite
Authored: Part III, pages 16-22 (7 pages) - Sections 14-20
Focus: Joint heterogeneous control, partial observability, HiDVFS, repository architecture, implemented state/actions/reward and local C checks
Owns: main.c, hardware.c, hardware.h, scheduler.c, scheduler.h, task_queue.c, task_queue.h, task.h, drl_scheduler.c, drl_scheduler.h, test_*.c, Makefile
References: [11]-[15]
Presents: 8:00-12:00 - Repository architecture, C engine and baseline schedulers
Reviews: Part IV (Atharva Sivaraman)
Indicative effort: about 28 hours

4. Atharva Sivaraman (24BEL0082)

Email: atharva.sivaraman2024@vitstudent.ac.in
Primary role: Evaluation, statistics, reproducibility and CI
Authored: Part IV, pages 23-29 (7 pages) - Sections 21-25
Focus: Revision priorities, evaluation plan, synthesis, extended review methodology, evidence grading and comparative evidence synthesis
Owns: benchmark_equal_work.py, compare_all_policies.c, compare_policies.c, results/equal_work_benchmark.json, .github/workflows/ci.yml
References: [16]-[20]
Presents: 12:00-16:00 - Evaluation design, statistics and the work-normalised results
Reviews: Part V (Yash Jain)
Indicative effort: about 28 hours

5. Yash Jain (24BEL0035)

Email: yash.jain2024a@vitstudent.ac.in
Primary role: Documentation, dashboard and document editor
Authored: Part V, pages 30-35 (6 pages) - Sections 26-29; Appendices A and B
Focus: Remediation and re-measurement, statistical protocol, team governance, limitations and ethics, artefact manifest and glossary
Owns: README.md, TEAM.md, CONTRIBUTION.md, CHANGELOG.md, LICENSE, docs/, app.py
References: [21]-[25]
Presents: 16:00-20:00 - Remediation, claim boundaries, limitations and conclusion
Reviews: Part I (Aditya Rajeshwar Karn)
Indicative effort: about 26 hours

9. Declaration

We certify that the work reported in the accompanying literature review and repository was carried out by the team members named above, that each member made a substantial contribution to the parts assigned to them, and that the division recorded here accurately reflects the way the work was shared. All five members have read and approved the final manuscript, the corrected benchmark results and the released code. Simulated energy figures are described as simulated throughout, and no measured hardware saving is claimed.

Name	Reg. No.	Signature	Date
Aditya Rajeshwar Karn	24BEL0087		
Yuvraj Singh	24BEL0057		
Aditya Singh	24BEL0008		
Atharva Sivaraman	24BEL0082		
Yash Jain	24BEL0035		

Note: This allocation is a working record. Each member should confirm their row and sign before submission. If the actual division differed, edit this document and the corresponding matrix in Section 28.2 of the review so that both records agree.